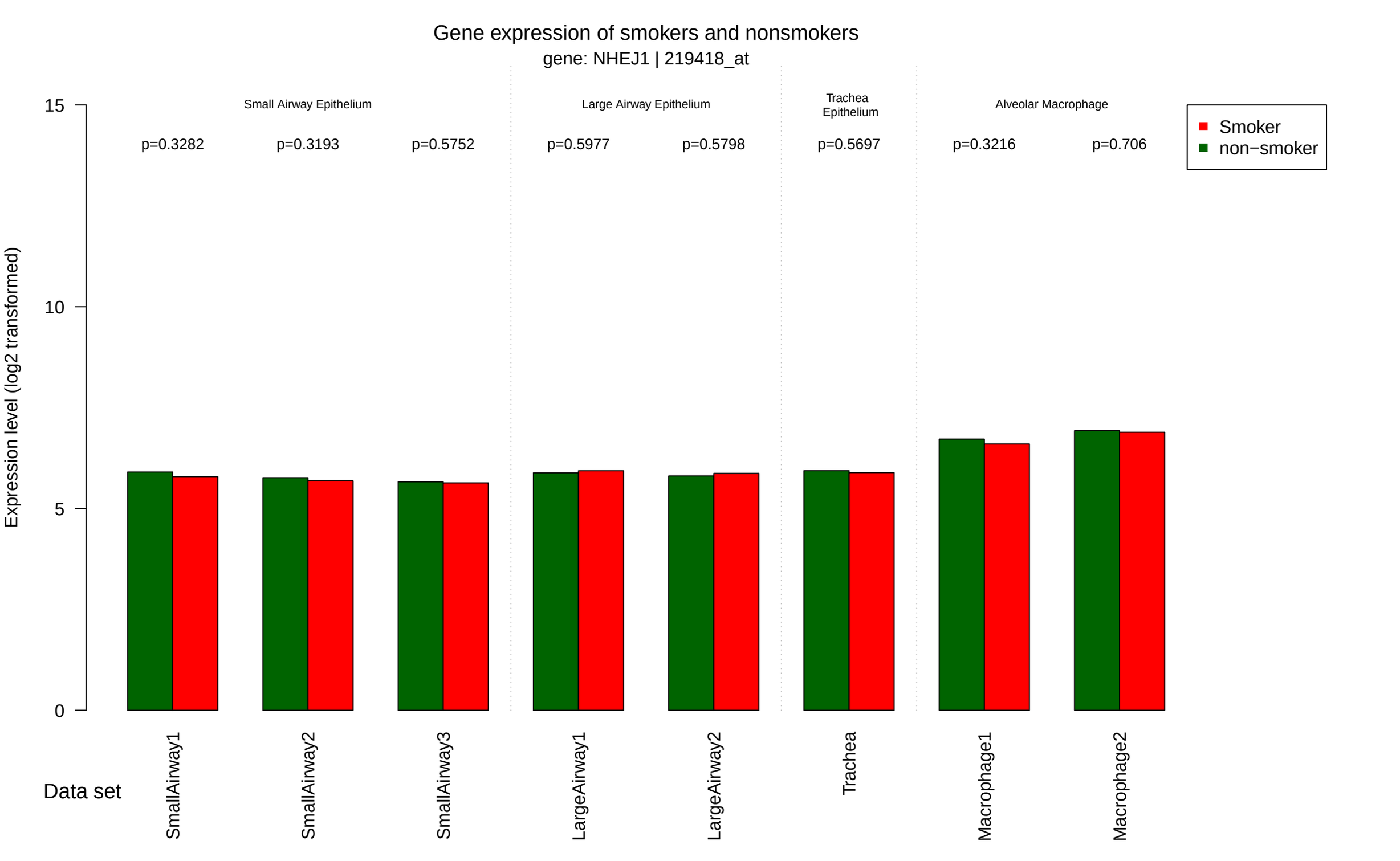
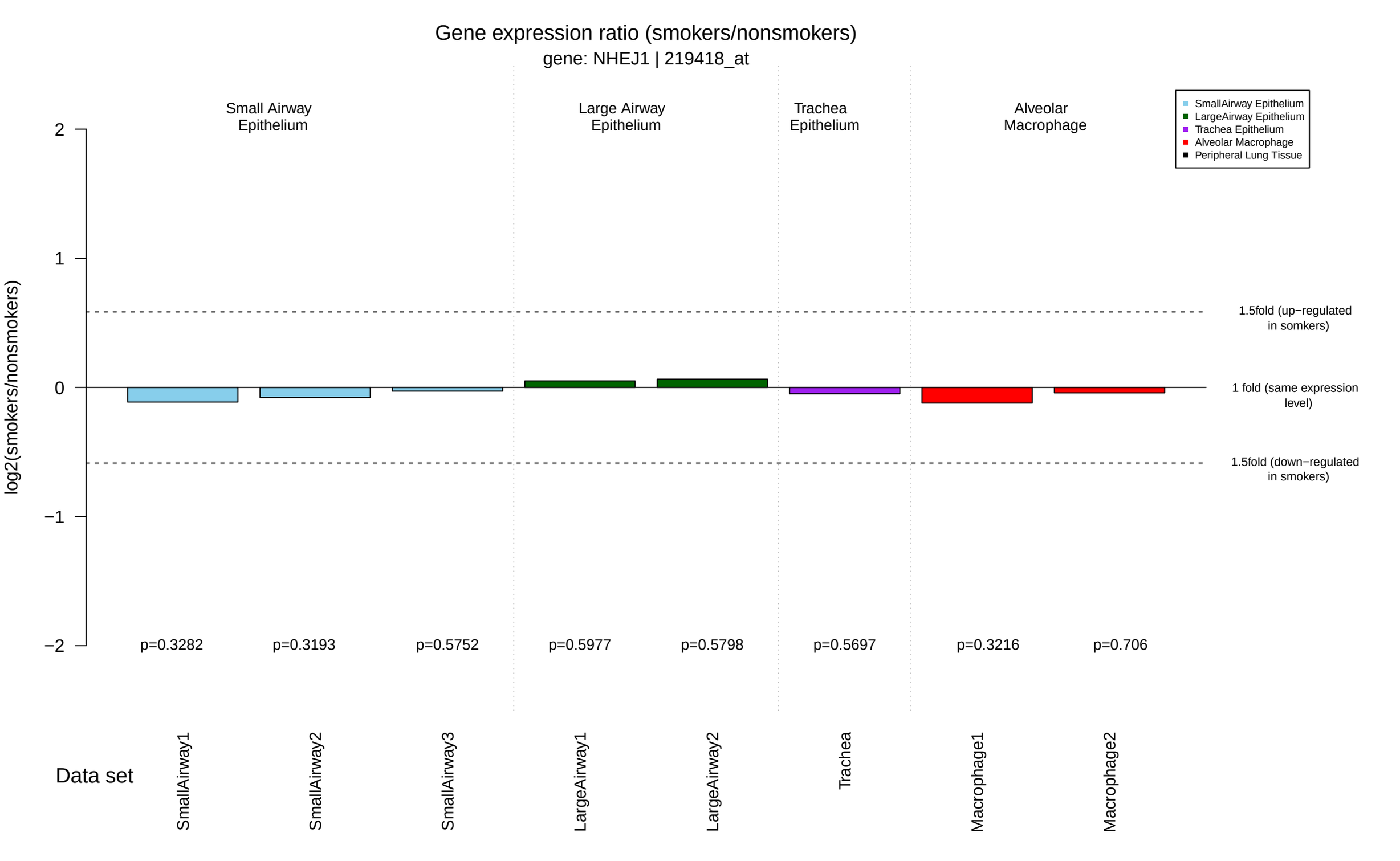
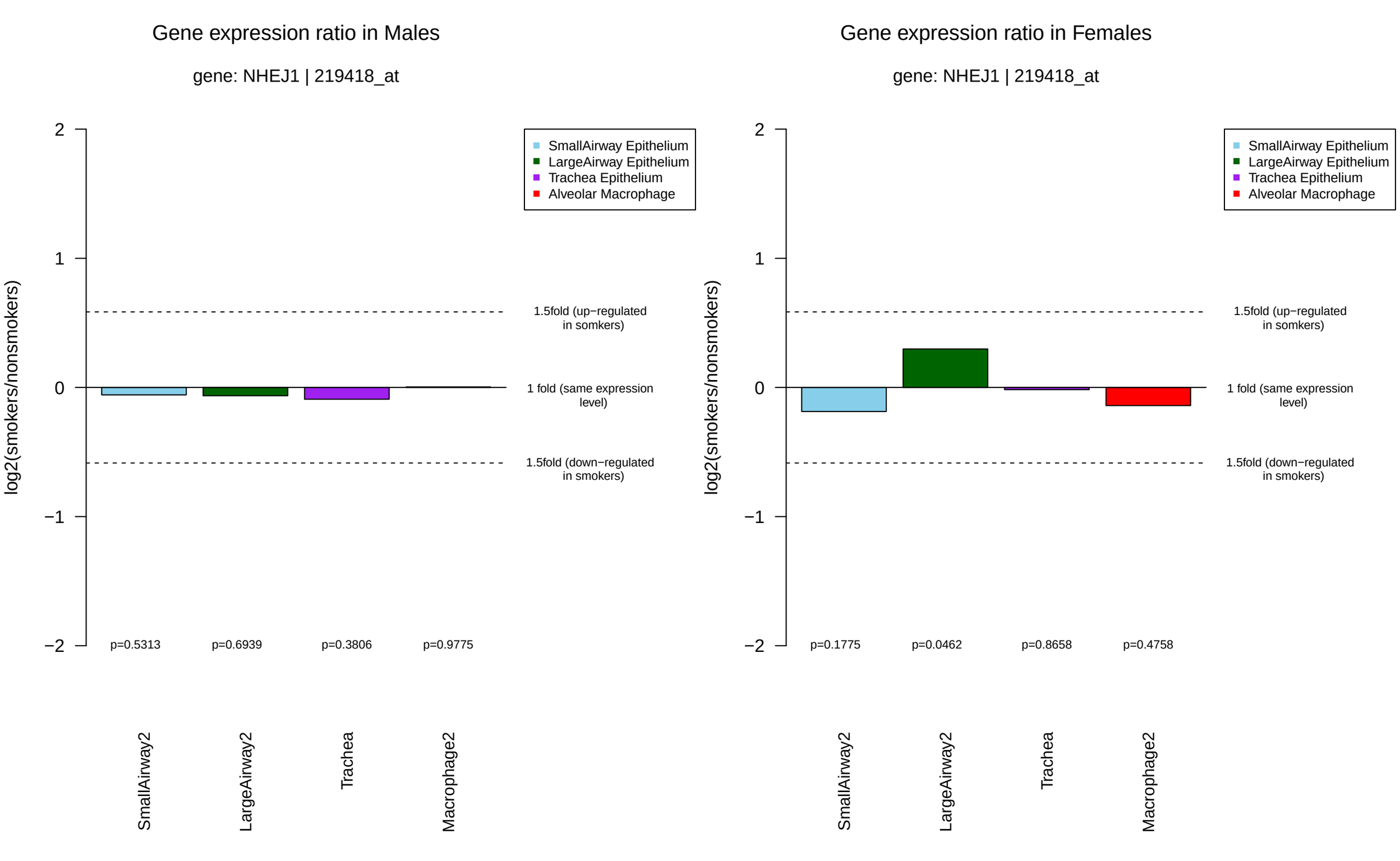
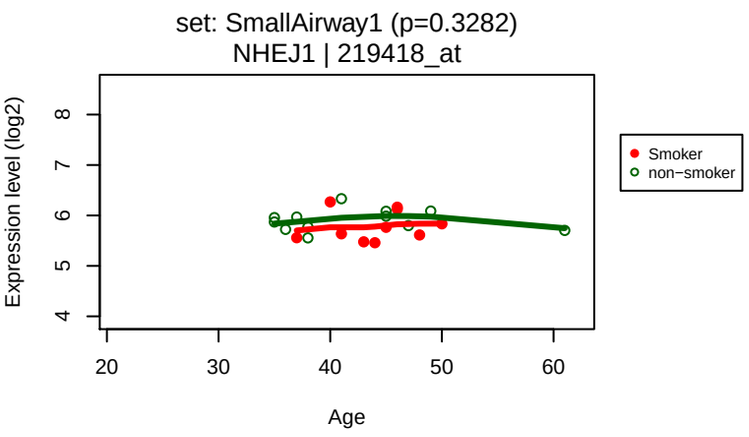
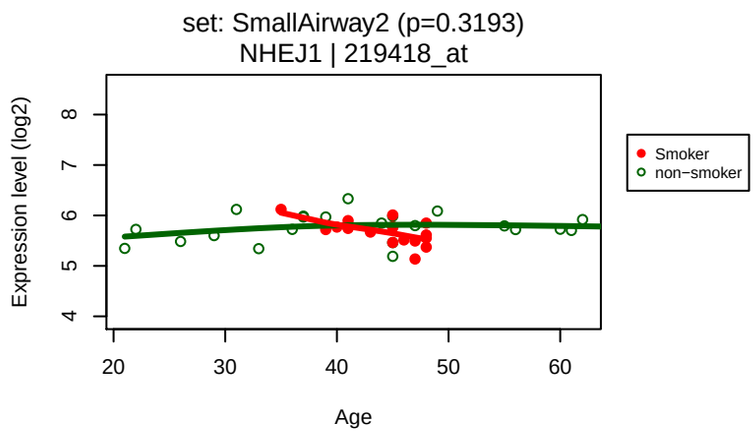
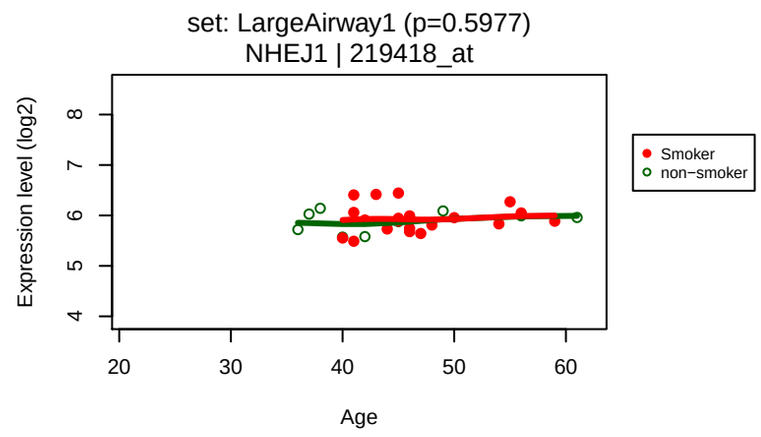

Gene expression of smokers and nonsmokers
gene: NHEJ1 | 219418_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3282
p=0.3193
p=0.5752
p=0.5977
p=0.5798
p=0.5697
p=0.3216
p=0.706
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

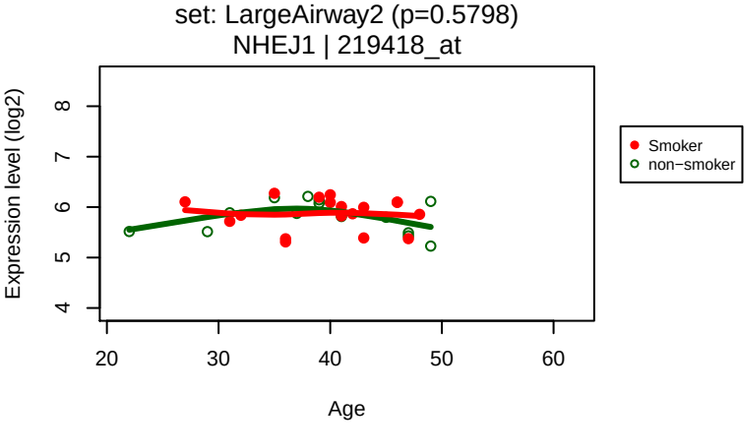

Gene expression ratio (smokers/nonsmokers)
gene: NHEJ1 | 219418_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3282
p=0.3193
p=0.5752
p=0.5977
p=0.5798
p=0.5697
p=0.3216
p=0.706
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

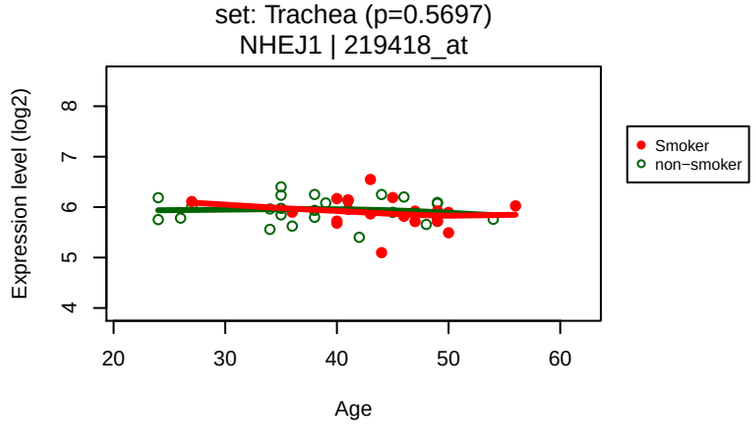

Gene expression ratio in Males
gene: NHEJ1 | 219418_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5313
p=0.6939
p=0.3806
p=0.9775
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NHEJ1 | 219418_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1775
p=0.0462
p=0.8658
p=0.4758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

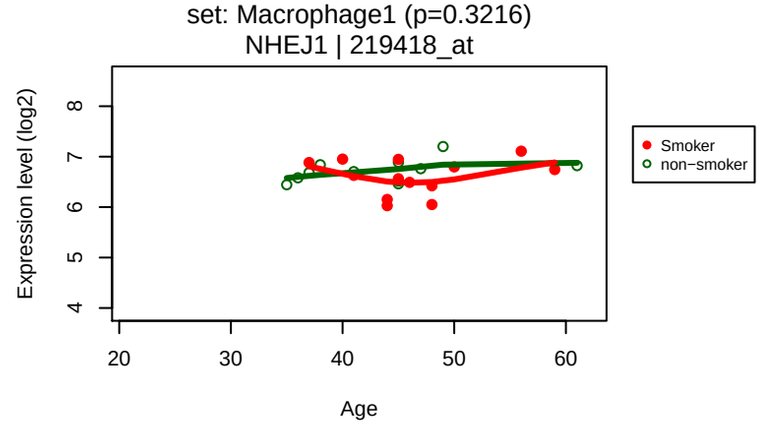

set: SmallAirway1 (p=0.3282)
NHEJ1 | 219418_at
Expression level (log2)
Age
Smoker
non−smoker

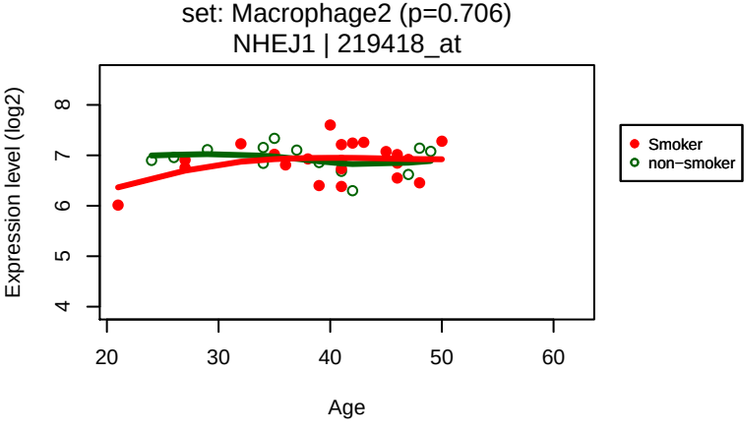

set: SmallAirway2 (p=0.3193)
NHEJ1 | 219418_at
Expression level (log2)
Age
Smoker
non−smoker

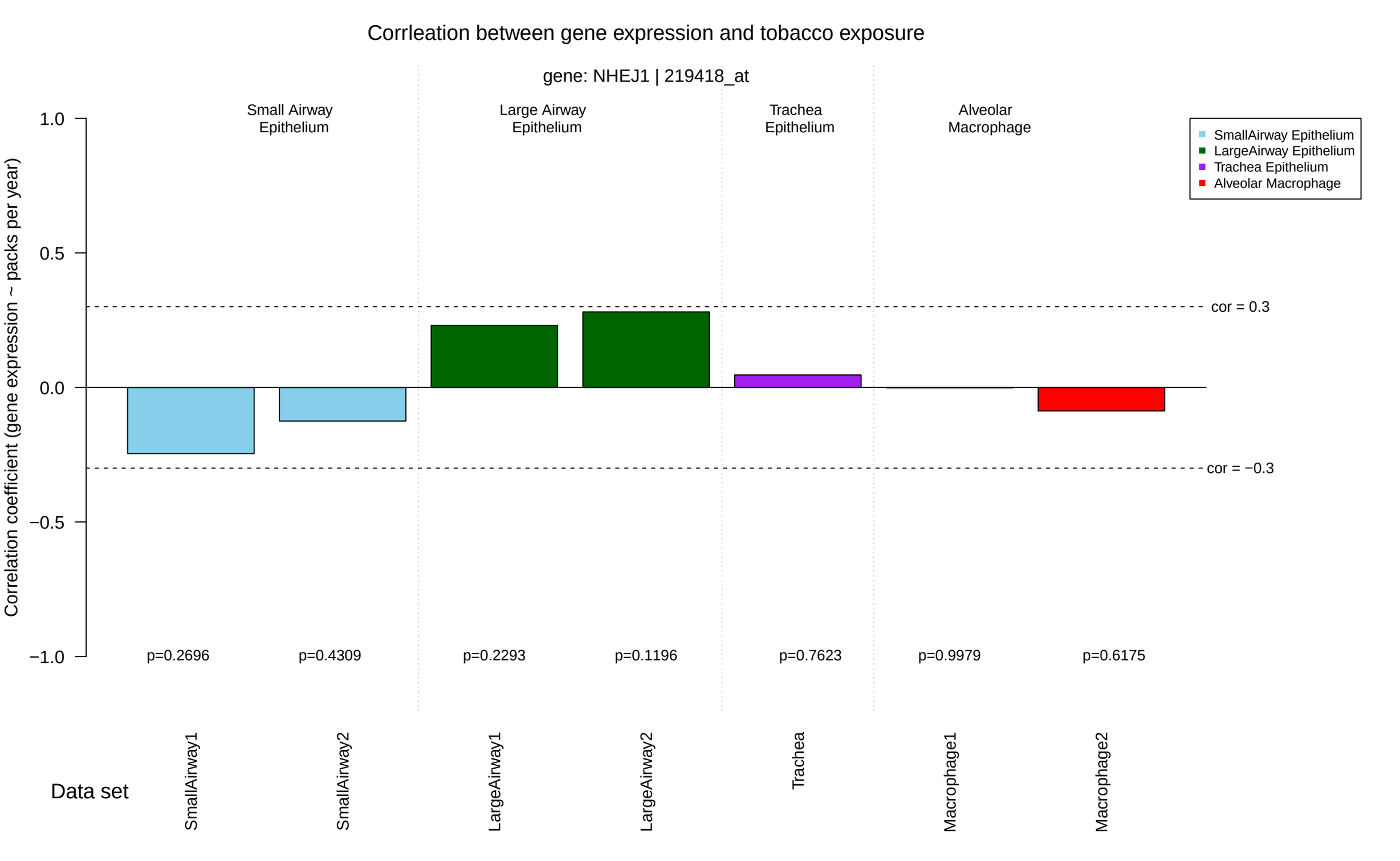

set: LargeAirway1 (p=0.5977)
NHEJ1 | 219418_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5798)
NHEJ1 | 219418_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5697)
NHEJ1 | 219418_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3216)
NHEJ1 | 219418_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.706)
NHEJ1 | 219418_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NHEJ1 | 219418_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2696
p=0.4309
p=0.2293
p=0.1196
p=0.7623
p=0.9979
p=0.6175
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2